

भारतीय मानक

IS 875 (Part 1) : 2026

Indian Standard

**इमारतों और संरचनात्मक ढांचों के लिए
भारतीय मानक डिज़ाइन भार (भूकंप के
अलावा) — रीति संहिता**

**भाग 1 स्थिर भार — भवन, निर्माण और संग्रहित
सामग्री का इकाई भार**

(तीसरा पुनरीक्षण)

**Design Loads (Other than
Earthquake) for Buildings and
Structures — Code of Practice**

**Part 1 Dead Loads — Unit Weights of
Building, Construction and Stored
Materials**

(Third Revision)

ICS 91.100; 91.080.01

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FOREWORD

This Indian Standard (Part 1) (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Structural Safety Sectional Committee had been approved by the Civil Engineering Division Council.

A building has to perform many functions satisfactorily. Amongst these functions are the utility of the building for the intended use and occupancy, structural safety, fire safety; and compliance with hygienic sanitation, ventilation and daylight standards. The design of the building is dependent upon the minimum requirements prescribed for each of the above functions. The minimum requirements pertaining to the structural safety of the buildings are being covered in the code by way of laying down minimum design loads which have to be assumed for dead loads, imposed loads, snow loads, and other external loads, the structure would be required to bear. Strict conformity to loading standards recommended in this code, it is hoped, will not only ensure the structural safety of the buildings which are being designed and constructed in the country and thereby reduce the hazards to life and property caused by unsafe structure, but also eliminate the waste caused by assuming unnecessary heavy loadings.

This standard deals with dead loads to be considered in the design of buildings and structures and this is given in the form of unit weight of materials. The unit weight of other materials that are likely to be stored in a structure are also included for the purpose of determining dead loads due to stored materials.

This Indian Standard was first published in 1957 for the guidance of civil engineers, designers and architects associated with planning and design of buildings to determine loads acting on building and structures. It included the provisions for the basic design loads (dead loads, live loads, wind loads and seismic loads) to be considered in the design of buildings. In its first revision in 1964, the wind pressure provisions were modified on the basis of studies of wind phenomenon and its effect on structures, undertaken by the special committee in consultation with the Indian Meteorological Department. In addition to this, new clauses on wind loads for butterfly type structures were included; wind pressure coefficients for sheeted roofs both curved and sloping, were modified; seismic load provisions were deleted (separate code having been prepared) and metric system of weights and measurements was adopted.

With the increased adoption of the code, a number of comments were received on provisions on live load values adopted for different occupancies. Simultaneously, live load surveys have been carried out in America and Canada to arrive at realistic live loads based on actual determination of loading (movable and immovable) in different occupancies. Keeping this in view and other developments in the field of wind engineering, the Sectional Committee responsible for the preparation of the standard prepared in 1987 the second revision in the following five parts:

Part 1 Dead loads

Part 2 Imposed loads

Part 3 Wind loads

Part 4 Snow loads

Part 5 Special loads and loads combinations

It was further revised in 1987, and IS 1911 : 1967 'Schedule of unit weights of building materials (*second revision*)' was incorporated in the standard.

Earthquake load are covered in separate standards, namely IS 1893 (Part 1 to 4) 'Criteria for earthquake resistant design of structures' which should be considered along with the above loads.

Since the publication of the second revision in 1987, there have been significant improvements in material technology leading to use of a number of new and improved materials for construction of buildings and structures. In this current revision, unit weights of many common new materials have been incorporated, and data on existing materials were reviewed and revised as necessary. The revisions of the corresponding Indian standards of the building materials have

(Continued on third cover)

Indian Standard

**DESIGN LOADS (OTHER THAN EARTHQUAKE) FOR
BUILDINGS AND STRUCTURES — CODE OF PRACTICE**

**PART 1 DEAD LOADS — UNIT WEIGHTS OF BUILDING,
CONSTRUCTION AND STORED MATERIALS**

(Third Revision)

1 SCOPE

1.1 This code (Part 1) covers unit weight of materials, and parts or components in a structure that apply to determination of dead loads in the design of structures. The unit weight of materials that are likely to be stored in a structure are also specified for the purpose of load calculations along with angles of material friction as appropriate.

1.2 The unit weight of materials is given in kN. The equivalent mass in kg may be determined by multiplying the weight by 1 000 and dividing by acceleration due to gravity, 9.81 m/s².

1.3 In this code, the words ‘structure’ and ‘building’ interchangeably refer to buildings and all other types of structures.

NOTE — [Table 1](#) gives the unit weight of individual building materials in alphabetical order.

2 REFERENCES

The standards listed in [Annex A](#) contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

3 DEAD LOAD

3.1 Assessment of Dead Load

The dead load in a building shall comprise the weight of all walls, partitions, floors and roofs, and shall include the weights of all other permanent constructions (including built-in partitions, finishes, cladding and other similarly incorporated architectural and structural items, weight of fixed service equipment) in the building.

3.2 The unit weight of materials used in construction of buildings and structures are specified in [Table 1](#). Also mentioned alongside are the corresponding Indian Standards, if applicable.

3.3 Gardening and Landscaping

Elements on roof gardening or landscaping, including soil, plants, and drainage layers, along with other features like walkways, fences, and walls, are fixed in place and thus classified as dead loads. While the weight of walkways, fences, and walls remains constant. The weight of soil and drainage layers, which support plant growth, can vary significantly due to their ability to absorb and retain water. For load calculations, when the weight adds to other structural loads, the dead load should be based on fully saturated soil and drainage layers to account for maximum weight. If the weight is used to counteract uplift forces, calculations should assume the dry weight of soil and drainage layers. Vegetative or landscaped roof areas may retain even more water than fully saturated soil, either from rainfall or irrigation. This additional water retention should be included in the design to accurately assess load variations. In the design calculation of overturning, uplifting and sliding effects, the beneficial effects of soil shall not be considered. When the soil is used as dead weight and locked permanently, the same may be considered.

4 STORED AND MISCELLANEOUS MATERIALS

4.1 Units weights of stored and miscellaneous materials intended for dead load calculation and other general purposes are given in [Table 2](#).

4.2 [Table 2](#) gives unit weight of stored materials, and the table shall not be used for volume calculations or any other similar purposes.

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Table 1 Unit Weight of Building Materials

(Clauses 1.3 and 3.2)

Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
1.	Acoustical Material (see IS 2526)			
	Eelgrass	10	$0.57 \text{ to } 0.765 \times 10^{-2}$	m ²
	Glass fibre, min	10	0.38×10^{-2}	m ²
	Hair	10	1.91×10^{-2}	m ²
	Mineral wool	10	1.345×10^{-2}	m ²
	Slag wool		2.65	m ³
	Cork		2.35	m ³
	Wood fibre board		5.9 to 6.15	m ³
	Wood particle board			
	As per IS 312		3.92	m ³
	As per IS 1282		4.9 to 9.9	m ³
	As per IS 3478		8.9 to 11.8	m ³
	Compressed wood wool	—	3.95 to 4.45	m ³
	Mineral/glass wool quilts and mats	—	0.157 to 0.314	m ³
	Mineral compressed glass wool tiles	—	1.87 to 2.06	m ³
	Polyester board	—	1.87 to 2.06	m ³
2.	Aggregate, Coarse			
	Broken stone ballast			
	Dry, well-shaken	—	15.70 to 18.35	m ³
	Perfectly wet	—	18.85 to 21.95	m ³
	Shingles, 3 mm to 38 mm	—	14.35	m ³
	Broken bricks:			
	Fine	—	14.20	m ³
	Coarse	—	9.90	m ³
	Foam slag (foundry pumice)	—	6.85	m ³
	Cinder	—	7.85	m ³
	Slag			
	Iron slag		32.4	m ³
	Steel Slag		31 to 36	m ³
	Copper slag		31 to 36	m ³
	River pebbles		15	m ³
3.	Aggregate, Fine			
	Sand			
	Dry, clean	—	15.10 to 15.70	m ³
	River	—	18.05	m ³
	Wet	—	17.25 to 19.60	m ³
	Brick dust (surkhi)	—	9.90	m ³
	Crushed stone sand (see IS 383)		19 to 20	m ³
	Quarry dust		14 to 20	m ³
4.	Aggregate, Organic			
	Saw dust, loose	—	1.55	m ³
	Peat			
	Dry	—	5.50 to 6.30	m ³
	Sandy, compact	—	7.85	m ³
	Wet, compact	—	13.35	m ³

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Sl No.	Materials	Nominal Size or Thickness	Weight	
		mm	kN	per
(1)	(2)	(3)	(4)	(5)
5.	Asbestos			
	Felt	10	0.145	m ²
	Fibres:			
	Pressed	—	9.40	m ³
	Sprayed	10	0.02	m ²
	Natural	—	29.80	m ³
	Raw	—	5.90 to 8.85	m ²
6.	Asbestos Cement Building Pipes (see Sl No. 44 'pipes' in this Table)			
7.	Asbestos Cement Gutters [see IS 1626 (Part 2)]			
	Boundary wall gutters			
	510 mm × 150 mm × 255 mm	12.5	0.180	m
	455 mm × 150 mm × 305 mm	12.5	0.158	m
	305 mm × 150 mm × 225 mm	12.5	0.129	m
	280 mm × 125 mm × 180 mm	12.5	0.106	m
	Valley gutters			
	915 mm × 205 mm × 230 mm	12.5	0.250	m
	610 mm × 150 mm × 230 mm	12.5	0.162	m
	455 mm × 125 mm × 150 mm	12.5	0.146	m
	405 mm × 125 mm × 255 mm	12.5	0.131	m
	Half round gutters			
	150 mm	9.5	0.043	m
	230 mm	9.5	0.073	m
	305 mm	9.5	0.088	m
8.	Asbestos Cement Pressure Pipes (see Sl No 44 'Pipes' in this Table)			
9.	Asbestos Cement Sheetting (see IS 459)			
	Corrugated (pitch – 146 mm)	6	0.118 to 0.130	m ²
	Semi-corrugated (pitch 388 mm)	6	0.118 to 0.127	m ²
	Plain	5	0.09	m ²
10.	Bitumen	—	0.8-1	m ³
11	Binders			
	Fly ash loose		5.05 to 7.05	m ³
	Fly ash compacted		16.00	m ³
	Ground granulated blast furnace slag		14.9	m ³
	Silica fume		3.5 to 7	m ³
	Rice husk ash		1.0 to 3.5	m ³
	Metakaolin		8 to 10	m ³
12	Blocks			
	Lime-based solid blocks (see IS 3115), min	—	9.8	m ³
	Concrete blocks [see IS 2185 (Part 1)]			
	Grade A (hollow)	—	15	m ³
	Grade B (hollow)	—	11 to 15	m ³
	Grade C (Solid)	—	18	m ³
	Autoclaved cellular concrete blocks [see IS 2185 (Part 3)]		4.42 to 9.81	m ³

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
13(1)	Boards (2)	(3)	(4)	(5)
	Cork boards [see IS 4253 (Part 1 and Part 2)]			
	Compressed		2.16 to 3.73	m ³
	Ordinary		2.15 to 3.72	m ³
	Fibre building boards see IS 1658)			
	Medium hardboard	6, 8, 10, 12	3.43 to 7.84	m ³
	Standard hardboard	2,3,4,5,6,7	7.84 to 10.05	m ³
	Tempered hardboard	2.5, 3, 4, 5, 6	7.84 to 10.05	m ²
	Fibre cement boards (see IS 14862)		13 to 16	m ³
	Fibre insulation board, ordinary or flame-retardant type bitumen-bounded fibre insulation board (see IS 3348), <i>Max</i>	9, 12, 18, 25	3.924	M ³
	Plain gypsum plaster boards [see IS 2095 (Part 1)]			
	Type A	9.5, 12.5, 15	5.39	m3
	Type H1 and H2	9.5, 12.5, 15	5.88	m3
	Type D	9.5, 12.5, 15	7.84	m3
	Type F and FH	9.5, 12.5, 15	7.84	m3
	Type R1	9.5, 12.5, 15	8.33	m3
	Type R2	9.5, 12.5, 15	9.31	m3
	Type W	9.5, 12.5, 15	8.33	m3
	Coated/laminated gypsum plaster boards [see IS 2095 (Part 2)]	9.5, 12.5, 15	11.27	M ³
	Reinforced gypsum plaster boards and ceiling tiles [see IS 2095 (Part 3)], <i>min</i>			
	Fibre reinforced Gypsum		7.84	m ³
	Glass reinforced Gypsum plaster board		24.52	m ³
	Glass reinforced Gypsum plaster tiles		5.88	m ³
	Insulating board (<i>fibre</i>) (see IS 3348), <i>mean</i>		0.981	m ³
	Laminated board (<i>fibre</i>) Medium density (see IS 14587)		5.88 to 8.82	m ³
	Wood particle boards (see IS 3087)	–	4.90 to 8.85	m ³
	Low density wood particles boards for insulation purposes (see IS 3129)	–	3.52	m ³
	High density wood particle boards (see IS 3478)			
	Type 1 & 2, Grade A	–	11.30	m ³
	Type 1 & 2, Grade B	–	7.94	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
14.	Bricks			
	Common burnt clay bricks (see IS 1077)	—	15.70 to 18.85	m ³
	Engineering bricks	—	21.20	m ³
	Heavy duty bricks (see IS 2180)	—	24.50	m ³
	Pressed bricks	—	17.25 to 18.05	m ³
	Refractory bricks	—	17.25 to 19.60	m ³
	Sand cement bricks	—	18.05	m ³
	Sand lime bricks	—	20.40	m ³
	Pulverized fuel-ash lime bricks (see IS 12894)		12 to 15	m ³
	Pulverized fuel-ash cement bricks (see IS 16720)		10.8 to 19.6	m ³
	Acid resistance bricks (see IS 4860)		22.5 to 24.83	m ³
	Soil based bricks (see IS 1725)		17.5	m ³
	Perforated building bricks (see IS 2222)		6.8 to 7.75	m ³
	Calcium silicate bricks (see IS 4139)		25 to 29	m ³
15.	Brick Chips and Broken Bricks (see under Sl No. 2 'Broken Bricks' in this Table)			
16.	Brick dust (surkhi)	—	9.00	m ³
17.	Cast Iron, Manhole Covers (see IS 1726)			
	Double triangular (HD)	500	1.16	Cover
		560	1.37	Cover
	Circular (HD)	500	1.16	Cover
		560	1.37	Cover
	Circular (MD)	500	0.57	Cover
		560	0.63	Cover
	Rectangular (MD)	—	0.78	Cover
	Rectangular (LD) :			
	Single seal (Pattern 1)	—	0.23	Cover
	Single seal (Pattern 2)	—	0.15	Cover
	Double seal	—	0.28	Cover
	Square (LD):			
	Single seal	455	0.13	Cover
		610	0.25	Cover
	Double seal	455	0.23	Cover
		610	0.36	Cover
18.	Cast Iron, Manhole Frames (see IS 1726)			
	Double triangular (HD)	500	1.09	Frame
		560	1.13	Frame
	Circular (HD)	500	0.83	Frame
		560	1.06	Frame
	Circular (MD)	500	0.57	Frame
		560	0.63	Frame
	Rectangular (MD)	—	0.63	Frame
	Rectangular (LD):			
	Single seal (Pattern 1)	—	0.15	Frame
	Single seal (Pattern 2)	—	0.10	Frame
	Double seal	—	0.23	Frame

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Square (LD):			
	Single seal	455	0.07	Frame
		610	0.13	Frame
	Double seal	455	0.15	Frame
		610	0.18	Frame
19.	Cast Iron Pipes (see under SI No. 44 'Pipes' in this Table)			
20.	Cement			
	Ordinary (see IS 269) and aluminous (see 6452)	—	14.10	m ³
	Rapid-hardening (see IS 8041)	—	12.55	m ³
21.	Ceilings			
	Plaster on tile or concrete	13 mm thick	0.25	m ²
	Plaster on wood lath	25 mm thick	0.39	m ²
	Suspended metal lath and cement plaster	25 mm thick	0.74	m ²
	Suspended metal lath and gypsum plaster	25 mm thick	0.49	m ²
22.	Cement Concrete, Plain			
	Aerated		7.45	m ³
	No-fines, with heavy aggregate	—	15.70 to 18.80	m ³
	No-fines, with light aggregate	—	8.65 to 12.55	m ³
	With burnt clay aggregate	—	17.25 to 21.20	m ³
	With expanded clay aggregate	—	9.40 to 16.50	m ³
	With clinker aggregate	—	12.55 to 17.25	m ³
	With pumice aggregate	—	5.50 to 11.00	m ³
	With sand and gravel or crushed natural stone aggregate	—	22.00 to 23.50	m ³
	With saw dust	—	6.30 to 16.50	m ³
	With foamed slag aggregate	—	9.40 to 18.05	m ³
	With light weight cinder aggregate		6.4	m ³
	Light weight concrete		8 to 20	m ³
	Cellular concrete (Foam concrete) (see IS 6598)		3 to 5	m ³
	Grade A, max		3.13	m ³
	Grade B		3.13 to 3.92	m ³
	Grade C		3.93 to 4.9	m ³
	Transparent concrete		15	m ³
	Glass fiber reinforced concrete		18 to 22	m ³
	Alkali activated (Geopolymer) concrete (see IS 17452)		24 to 25	m ³
	Shotcrete (see IS 9012)		22 to 25	m ³
	Ferro cement		22 to 25	m ³
	Roller compacted concrete		22 to 25	m ³
	Controlled low strength mix		22 to 25	m ³
	Plum concrete (see IS 457)		22 to 25	m ³
	Autoclaved aerated concrete		4.42 to 9.81	m ³
	Vermiculite concrete		6 to 9	m ³

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
23.	Cement Concrete, Prestressed (conforming to IS 1343)	–	23.50	m ³
24.	Cement Concrete, Reinforced:			
	With 1 percent steel	–	22.75 to 24.20	m ³
	With 2 percent steel	–	23.25 to 24.80	m ³
	With 5 percent steel	–	24.80 to 26.50	m ³
25.	Damp-proofing (see SI No. 28 ‘Felt, bituminous, for waterproofing and damp proofing’ in this Table)			
26	Earth filling (see SI No. 49 ‘Soils and gravels in this Table)			
27.	Expanded Metal (conforming to IS 412)			
	Size of Mesh, Nominal			
	SWM	LWM		
	mm	mm		
	100	250	0.030	m ²
	100	250	0.024	m ²
	100	250	0.016	m ²
	75	200	0.042	m ²
	75	200	0.032	m ²
	75	200	0.021	m ²
	40	115	0.080	m ²
	40	115	0.060	m ²
	40	75	0.060	m ²
	40	75	0.028	m ²
	40	115	0.039	m ²
	40	75	0.039	m ²
	40	115	0.020	m ²
	40	75	0.020	m ²
	25	75	0.054	m ²
	25	75	0.038	m ²
	25	75	0.028	m ²
	25	75	0.021	m ²
	20	60	0.070	m ²
	20	50	0.070	m ²
	20	60	0.050	m ²
	20	50	0.050	m ²
	20	60	0.036	m ²
	20	50	0.036	m ²
	20	50	0.021	m ²
	20	60	0.021	m ²
	12.5	50	0.050	m ²
	12.5	50	0.050	m ²
	12.5	40	0.040	m ²
	12.5	50	0.030	m ²
	12.5	40	0.030	m ²
	12.5	50	0.025	m ²
	12.5	40	0.025	m ²
	10	40	0.050	m ²

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	10	40	0.035	m ²
	10	40	0.028	m ²
	95	285	0.050	m ²
	95	285	0.028	m ²
	95	285	0.020	m ²
	6	25	0.074	m ²
	6	25	0.048	m ²
	6	25	0.038	m ²
	5	20	0.050	m ²
	3	15	0.041	m ²
28.	Felt, Bituminous for Waterproof and Damp-proofing (see IS 1322)			
	Fibre base			
	Type 1 (under lay)	—	7.6×10^{-3}	m ²
	Type 2 (self-finished felt)	-	2.6×10^{-2}	m ²
	Hessian base			
	Type 3 (self-finished felt)			
	Grade 1	—	2.26×10^{-2}	m ²
	Grade 2	—	3.64×10^{-2}	m ²
29.	Finishing (see 'Floor finishes' given under Sl No. 31 'Flooring' and 46 'Roofing' in this Table)			
	Plaster:			
	Acoustic	10	0.08	m ²
	Anhydrite	10	0.21	m ²
	Barium Sulphate	10	0.28	m ²
	Fibrous	10	0.09	m ²
	Gypsum or lime	10	0.19	m ²
	Hydraulic lime or cement	10	0.23	m ²
	Plaster ceiling on wire netting	10	0.26	m ²
	NOTE — When wood or metal lathing is used, add	—	0.06	
30.	Float Glass (see IS 2835) and Safety Glass [see IS 2553 (Part 1)]			
	Sheet	2	0.049	m ²
		2.5	0.062	m ²
		3	0.074	m ²
		4	0.098	m ²
		5	0.123	m ²
		5.5	0.134	m ²
		6.5	0.167	m ²
		8	0.200	m ²
		10	0.250	m ²
		12	0.300	m ²
		15	0.375	m ²
		19	0.475	m ²

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
31.	Flooring			
	Asphalt flooring	10	0.22	m ²
	NOTE — For macadam finish, add	10	0.26	m ²
	Compressed cork	10	0.04	m ²
	Floors, structural:			
	Hollow clay blocks including reinforcement and mortar jointing between blocks, but excluding any concrete topping	100	1.47	m ²
		125	1.67	m ²
		150	1.86	m ²
		175	2.16	m ²
		200	2.55	m ²
	NOTE — Add extra for concrete topping			
	Hollow clay blocks including reinforcement and concrete ribs between blocks, but excluding any concrete topping	100	1.18	m ²
		115	1.27	m ²
		125	1.37	m ²
		140	1.47	m ²
		150	1.57	m ²
		175	1.76	m ²
		200	1.96	m ²
	NOTE — Add extra for concrete typing			
	Hollow concrete units including any concrete topping necessary for constructional purposes	100	1.67	m ²
		125	1.96	m ²
		150	2.16	m ²
		175	2.35	m ²
		200	2.65	m ²
		230	3.14	m ²
	Floors, wood:			
	Hard wood	22	0.16	m ²
		28	0.20	m ²
	Soft wood	22	0.11	m ²
		28	0.13	m ²
	Weight of mastic used in laying wood block flooring	—	0.015	m ²
	NOTE — All thicknesses are 'finished thicknesses'.			
	Floor finishes:			
	Clay floor tiles (see IS 1478)	12.5 to 25.4	0.10 to 0.2	m ²
	NOTE — This weight is 'as laid' but excludes screeding.			
	Magnesium oxychloride:			
	Normal type (saw dust filler)	10	0.142	m ²
	Heavy duty type (mineral filler)	10	0.216	m ²
	Parquet flooring	—	0.08 to 0.12	m ²
	Rubber (see IS 809)	3.2	0.048 to 0.062	m ²
		4.8	0.070 to 0.09	m ²
		6.4	0.093 to 0.130	m ²
	Terra cotta, filled 'as laid'	—	5.54 to 7.06	m ²
	Terrazzo paving 'as laid'	10	0.23	m ²
32.	Foam Slag, Foundry Pumice	—	6.85	m ³
33.	Gutters, Asbestos Cement (see under 7 'Asbestos cement gutter' in this Table)			

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
34.	Gypsum			
	Gypsum mortar [see IS 2547 (Part 1 and Part 2)]	—	18.6	m ³
	Gypsum powder (see IS 12679)	—	13.89 to 17.25	m ³
35.	Iron			
	Pig	—	70.60	m ³
	Gray, cast	—	68.95 to 69.90	m ³
	White, cast	—	74.30 to 75.70	m ³
	Wrought	—	75.70	m ³
36.	Lime			
	Lime concrete with burn clay Aggregate (see IS 2541)	—	18.80	m ³
	Lime mortar	—	15.70 to 18.05	m ³
	Lime plaster (see IS 2394)	—	17.25	m ³
	Lime stored in lumps, uncalcined	—	12.55 to 14.10	m ³
	Lime, unslaked, freshly burnt in pieces	—	8.60 to 10.20	m ³
	Lime slaked, fresh	—	5.70 to 6.30	m ³
	Lime slaked, after 10 days	—	7.85	m ³
	Lime, unslaked (<i>kankar</i>)	—	11.55	m ³
	Lime, slaked (<i>kankar</i>)	—	10.00	m ³
37.	Linoleum (see IS 653)			
	Sheets and tiles	4.5	0.0569	m ²
		3.2	0.0402	m ²
		2.0	0.0265	m ²
		1.65	0.0215	m ²
38.	Masonry			
	Brick Masonry (excluding plaster)			
	With common burnt clay bricks (see IS 1077)	—	19.25	m ³
	Engineering bricks	—	21.2	m ³
	Glazed bricks	—	20.40	m ³
	Pressed bricks	—	18.65	m ³
	Block Masonry			
	With lime based solid blocks		13.55	m ³
	Hollow concrete blocks			
	Grade A		15.4	m ³
	Grade B		12.00	m ³
	Solid Concrete Blocks		18.15	m ³
	Autoclaved cellular concrete blocks		9.1 to 11.1	m ³
39.	Masonry, Stone			
	Cast	—	21.75	m ³
	Dry rubble	—	20.40	m ³
	Granite ashlar	—	25.00	m ³
	Granite rubble	—	23.55	m ³
	Lime stone ashlar	—	25.10	m ³
	Marble dressed	—	26.50	m ³
	Sand stone	—	22.00	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
40.	Mastic Asphalt	10	0.216	m ²
41.	Metal Sheet, Protected Galvanized Steel Sheets, Plain (see IS 277)			
	Class A (corrugated)	1.60	0.131	m ²
		1.00	0.104	m ²
		0.9	0.084	m ²
		0.80	0.069	m ²
		0.63	0.056	m ²
	Class B (corrugated)	1.60	0.129	m ²
		1.00	0.102	m ²
		0.9	0.083	m ²
		0.80	0.067	m ²
		0.63	0.054	m ²
	Class C (corrugated)	1.60	0.128	m ²
		1.00	0.101	m ²
		0.9	0.081	m ²
		0.80	0.066	m ²
		0.63	0.053	m ²
	Class D (corrugated)	1.60	0.127	m ²
		1.00	0.100	m ²
		0.9	0.081	m ²
		0.80	0.065	m ²
		0.63	0.052	m ²
42.	Mortar			
	Cement	—	20.40	m ³
	Gypsum	—	11.75	m ³
	Lime	—	15.70 to 18.05	m ³
43.	Pipes			
	Asbestos cement pipes [see IS 1626 (Part 1)]	50	0.032 to 0.034	m
		60	0.032 to 0.043	m
		80	0.051 to 0.054	m
		90	0.052 to 0.060	m
		100	0.058 to 0.065	m
		125	0.072 to 0.086	m
		150	0.086 to 0.108	m
	Asbestos cement pressure pipes [(see IS 1592)]	50	0.056	m
		80	0.067	m
		100	0.090	m
		125	0.139	m
		150	0.175	m
		200	0.264	m
		250	0.380	m
		300	0.539	m
	Cast iron pipes (see IS 1729			m ³
	Standard overall length	50	0.073	pipe
	1.8 m with socket	75	0.108	Pipe

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
		100	0.137	pipe
		125	0.196	pipe
		150	0.255	pipe
	Standard overall length 1.5 m with socket	50	0.064	pipe
		75	0.093	pipe
		100	0.123	pipe
		125	0.172	pipe
		150	0.230	Pipe
	Pressure pipes for water, gas and sewage:			
	a) Centrifugally cast (see IS 1536)			
	Socket and spigot pipes			
	Barrel:			
	Class LA	80	0.145	m
		100	0.182	m
		125	0.237	m
		150	0.297	m
		200	0.432	m
		250	0.582	m
		300	0.750	m
		350	0.944	m
		400	1.146	m
		450	1.383	m
		500	1.620	m
		600	2.156	m
		700	2.718	m
		750	3.111	m
	Class A	80	0.151	m
		100	0.201	m
		125	0.259	m
		150	0.326	m
		200	0.472	m
		250	0.637	m
		300	0.824	m
		350	1.030	m
		400	1.262	m
		450	1.530	m
		500	1.775	m
		600	2.367	m
		700	3.056	m
		750	3.422	m
	Class B	80	0.172	m
		100	0.216	m
		125	0.281	m
		150	0.352	m
		200	0.511	m
		250	0.692	m
		300	0.896	m
		350	1.112	m
		400	1.368	m
		450	1.657	M

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
		500	1.929	m
		600	2.578	m
		700	3.317	m
		750	3.733	m
	Sockets for Class LA, Class A and Class B barrels	80	0.054	Socket
		100	0.069	Socket
		125	0.090	Socket
		150	0.113	Socket
		200	0.165	Socket
		250	0.225	Socket
		300	0.292	Socket
		350	0.368	Socket
		400	0.454	Socket
		450	0.549	Socket
		500	0.647	Socket
		600	0.876	Socket
		700	1.145	Socket
		750	1.292	Socket
	Flanged pipe with screwed flanges:			
	Barrel:			
	Class A	80 to 300	Same as for centrifugally cast socket and spigot piles, Class A	
	Class B	80 to 300	Same as for centrifugally cast socket and spigot piles, Class B	
	Flanges for Class A and Class B barrels	80	0.042	Flange
		100	0.049	Flange
		125	0.065	Flange
		150	0.080	Flange
		200	0.112	Flange
		250	0.144	Flange
		300	0.182	Flange
	Vertically cast socket and spigot pipes (see IS 1537)			
	Barrel: Class A	80 to 750	Same as for centrifugally cast socket and spigot pipes, Class A	
		800	3.82	m
		900	4.65	m
		1 000	5.59	m
		1 100	6.59	m
		1 200	7.67	m
		1 500	11.98	m
	Class B	80 to 750	—	—
		800	—	m
		900	5.07	m
		1 000	6.07	m
		1 100	7.23	m
		1 200	8.35	m
		1 500	13.07	m

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Socket of Class A and Class B barrels	80 to 750	—	—
		800	—	Socket
		900	179	Socket
		1 000	218	Socket
		1 100	260	Socket
		1 200	360	Socket
		1 500	491	Socket
	Sand cast (flanked pipes)			
	Barrel:			
	Class A	80 to 750	Same as for centrifugally cast socket and spigot pipes, Class A	
		800 to 1 500	—	
	Class B	80 to 750	Same as for vertically cast socket and spigot pipes, Class B	
		800 to 1 500		
	Flanges for Class A and Class B barrels	80	0.036	Flange
		100	0.041	Flange
		125	0.052	Flange
		150	0.066	Flange
		200	0.091	Flange
		250	0.117	Flange
		300	0.145	Flange
		350	0.186	Flange
		400	0.229	Flange
		450	0.250	Flange
		500	0.315	Flange
		600	0.431	Flange
		750	0.587	Flange
		700	0.685	Flange
		800	0.792	Flange
		900	0.928	Flange
		1 000	1.18	Flange
		1 100	1.38	Flange
		1 200	1.70	Flange
		1 500	2.71	Flange
	Concrete pipes (<i>see</i> IS 458)			
	Class NP 1 (unreinforced non- pressure pipes)	80	0.19	m
		100	0.22	m
		150	0.30	m
		250	0.40	m
		300	0.69	m
		350	0.84	m
		400	0.95	m
		450	1.17	m
	Class NP 2 (reinforced concrete, light duty, non- pressure pipes)	80	0.196	m
		100	0.215	m
		150	0.324	m
		250	0.510	M

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
		300	0.736	m
		350	0.902	m
		400	1.02	m
		450	1.26	m
		500	1.38	m
		600	1.89	m
		700	2.19	m
		800	2.81	m
		900	3.51	m
		1 000	4.30	m
		1 100	5.15	m
		1 200	6.09	m
		1 400	8.18	m
		1 600	9.93	m
		1 800	12.58	m
	Class NP 3 (reinforced concrete, heavy duty, non-pressure pipes)	350	2.35	m
		400	2.63	m
		450	2.91	m
		500	3.19	m
		600	4.02	m
		700	4.61	m
		800	5.92	m
		900	7.39	m
		1 000	8.13	m
		1 100	10.34	m
		1 200	11.18	m
	Class P1 (reinforced concrete pressure pipes safe for 20 MPa pressure tests)	80	0.196	m
		100	0.235	m
		150	0.324	m
		250	0.510	m
		300	0.736	m
		350	0.902	m
		400	1.02	m
		450	1.26	m
		500	1.38	m
		600	1.89	m
		700	2.19	m
		800	2.81	m
		900	3.51	m
		1 000	4.30	m
		1 100	5.15	m
		1 200	6.09	m
	Class P2 (reinforced concrete pressure pipes safe for 40 MPa pressure tests)	80	0.196	m
		100	0.235	m
		150	0.324	m
		250	0.608	m
		300	1.01	m
		350	1.31	m
		400	1.67	m

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
		450	1.84	m
		500	1.56	m
		600	3.20	m
	Class P3 (reinforced concrete pressure pipes safe for 60 MPa pressure tests)	80	0.196	m
		100	0.235	m
		150	0.324	m
		250	0.736	m
		300	1.15	m
		350	1.65	m
		400	2.04	m
	Lead pipes [see IS 404 (Part 1)] (service and distribution pipes to be laid underground) :			
	For working 40 MPa	10	0.018	m
		13	0.031	m
		20	0.042	m
		25	0.060	m
		30	0.074	m
		40	0.091	m
		50	0.142	m
	For working 70 MPa	10	0.022	m
		15	0.038	m
		20	0.050	m
		25	0.069	m
		32	0.126	m
	For working 100 MPa	40	0.175	m
		10	0.029	m
		15	0.048	m
		20	0.067	m
		25	0.105	m
	Service pipes to be fixed or laid above ground:			
	For working pressure 40 MPa	10	0.014	m
		15	0.021	m
		20	0.027	m
		25	0.036	m
		32	0.059	m
		40	0.091	m
		50	0.142	m
	For working pressure 70 MPa	10	0.018	m
		15	0.024	m
		20	0.030	m
		25	0.069	m
		32	0.126	m
	For working pressure 100 MPa	40	0.175	m
		10	0.029	m
		15	0.048	m
		20	0.067	m
		25	0.105	M

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Cold water distribution pipes to be fixed or laid above ground:			
	For working pressure 25 MPa	10	0.014	m
		15	0.021	m
		20	0.027	m
		25	0.036	m
		32	0.048	m
		40	0.067	m
		50	0.084	m
	For working pressure 40 MPa	10	0.014	m
		15	0.021	m
		20	0.027	m
		25	0.036	m
		32	0.059	m
		40	0.091	m
		50	0.142	m
	Hot water distribution pipes to be fixed or laid above ground:			
	For working pressure 20 MPa	10	0.015	m
		15	0.023	m
		20	0.031	m
		25	0.041	m
		32	0.062	m
		40	0.082	m
		50	0.142	m
	For working pressure 35 MPa	10	0.015	m
		15	0.027	m
		20	0.045	m
		25	0.085	m
		32	0.132	m
	Soil, waste, and soil and waste ventilation pipes	50	0.050	m
		75	0.073	m
		100	0.097	m
		150	0.160	m
	Flushing and warning pipes	20	0.020	m
		25	0.025	m
		32	0.032	m
		40	0.039	m
		50	0.049	m
	Gas pipes:			
	Heavy weight gas pipes	10	0.008	m
		15	0.017	m
		20	0.025	m
		25	0.034	m
		32	0.045	m
		40	0.061	m
		50	0.071	m
	Light weight gas pipes	10	0.008	m
		15	0.012	m
		20	0.020	m

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
		25	0.029	m
		32	0.037	m
		40	0.047	m
		50	0.058	m
	Stoneware, salt-glazed pipes (see IS 651)	100	0.137	m
		150	0.216	m
		200	0.324	m
		230	0.412	m
		250	0.510	m
		300	0.775	m
		350	0.980	m
		400	1.26	m
		450	1.44	m
		500	1.77	m
		600	2.35	m
44.	Plaster			
	Cement	10	0.204	m ²
	Lime	10	0.172	m ²
	Acoustic	10	0.078	m ²
	Anhydrite	10	0.206	m ²
	Barium sulphate	10	0.284	m ²
	Fibrous	10	0.088	m ²
	Gypsum	10	0.186	m ²
45.	Roofing			
	Asbestos cement sheeting			
	(see 'Asbestos cement sheeting' in Table 1)			
	Allahabad tiles (single) including battens (see Note below)	—	0.83	m ²
	Allahabad tiles (double) including battens (see Note below)	—	1.67	m ²
	Country tiles (single) including battens (see Note below)	—	0.69	m ²
	Country tiles (double) including battens (see Note below)	—	1.18	m ²
	Mangalore tiles battens (see Note below)	—	0.64	m ²
	Mangalore tiles bedded in mortar over flat tiles (see Note below)	—	1.08	m ²
	Mangalore tiles with flat tiles (see Note below)	—	0.78	m ²
	Copper sheet roofing including laps and rolls	0.56	0.08	m ²
	Flat roofs:	0.72	0.10	m ²
	Clay tiles- hollow clay blocks/ clay floor tiles (see SI No. 31 'Flooring' in this Table)			
	Concrete hollow units (see SI No. 31 'Flooring' in this Table)			

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Galvanized iron sheeting (<i>see</i> Sl No. '41 Metal sheeting, protected' in this Table)			
	Glazed Roofing:			
	Glazing with Aluminum alloy bars for spans up to 3 m	6.4	0.19	m ²
	Glazing with lead-covering steel bars at 0.6 m centres	6.4	0.25 to 0.28	m ²
	States on battens	—	0.34 to 0.49	m ²
	Thatch with battens	—	0.34 to 0.49	m ²
	NOTE — Weights acting vertically on horizontally projection to be multiplied by cosine of roof angle to obtain weights normal to the roof surface.			
	Roof finished			
	Bitumen macadam	10	0.22	m ²
	Felt roofing (<i>see</i> 28 'Felt, bituminous, for water-proofing and damp-proofing' in Table 1)	10	0.008	m ²
	Glass silk quilted	0.5	0.05	m ²
	Lead sheet	0.8	0.07	m ²
	Mortar screeding	10	0.21	m ²
46.	Sheeting			
	Asbestos (<i>see</i> under Sl No. 9 'asbestos cement sheeting' in this Table)			
	Galvanized iron (<i>see</i> under Sl No.39 'metal sheeting, protected' in this table)			
	Glass (<i>see</i> under Sl No.30 'Glass' in this table)			
	Plywood (<i>see</i> IS 303)	1	0.007	m ³
47.	Slugwool	—	2.65	m ³
48.	Soils and Gravels			
	Alluvial ground, undisturbed	—	15.69	m ³
	Broken stone ballast:			
	Dry, well-shaken	—	15.70 to 18.35	m ³
	Perfectly wet	—	18.85 to 21.95	m ³
	Chalk		15.70 to 18.85	m ³
	Clay:			
	China, compact	—	21.95	m ³
	Clay fills:			
	Dry, lumps	—	10.20	m ³
	Dry, compact	—	14.10	m ³
	Damp, compact	—	17.25	m ³
	Wet , compact	—	20.40	m ³
	Undisturbed	—	18.85	m ³
	Undisturbed, gravelly	—	20.40	m ³
	Crush stone sand	—	—	—
	Earth:			
	Dry	—	13.85 to 18.05	m ³
	Moist	—	15.70 to 19.60	m ³
	Gravel:			
	Loose	—	15.70	m ³
	Rammed	—	18.85 to 21.20	m ³
	Kaolin, compact	—	25.50	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Loam:			
	Dry, loose	—	11.75	m ³
	Dry, compact	—	15.70	m ³
	Wet, compact	—	18.85	m ³
	Loess, dry	—	14.10	m ³
	Marl, compact	—	17.25 to 18.85	m ³
	Mud, river, wet	—	17.25 to 18.85	m ³
	Peat:			
	Dry	—	5.50 to 6.30	m ³
	Sandy, compact	—	7.85	m ³
	Wet, compact	—	13.35	m ³
	Rip-rap	—	12.55 to 14.10	m ³
	Sand:			
	Dry, clean	—	15.10 to 15.70	m ³
	River	—	18.05	m ³
	Wet	—	17.25 to 19.60	m ²
	Shingles:			
	Aggregate 3 to 38 mm	—	13.75	m ²
	Fine sand:			
	Dry	—	15.70	m ²
	Saturated	—	20.40	m ²
	Silt, wet	—	17.25 to 18.85	m ²
49	Steel Sections			
	Density of steel sections shall be taken as 78.50 kg/m ³			
	Weight of hot rolled sections (ISJB, ISLB, ISMB, ISWB, ISNPB, ISWPB, ISSC, ISHB, ISJC, ISLC, ISMC, ISMPC, ISA, ISBPB) shall be referred from IS 808			
	Weight of hollow steel sections shall be referred from IS 4923			
	Weight of cold formed light gauge structural steel sections shall be referred from IS 811			
	Weight of rolled and slit T bars shall be referred from IS 1173			
	Weight of steel sheet piling sections shall be referred from IS 2314			
50.	Stone			
	Agate	—	25.50	m ³
	Aggregate	—	15.70 to 18.85	m ³
	Basalt	—	27.95 to 29.05	m ³
	Cast	—	21.95	m ³
	Chalk	—	21.50	m ³
	Dolomite	—	28.25	m ³
	Emery	—	39.25	m ³
	Flint	—	25.40	m ³
	Gneiss	—	23.55 to 26.40	m ³
	Granite	—	25.90 to 27.45	m ³
	Gravel:			
	Loose	—	15.70	m ³
	Moderately rammed, dry	—	18.85	m ³
	Green stone	—	28.25	m ³
	Gypsum	—	21.95 to 23.55	m ³
	Kota	—	—	—
	Laterite	—	20.40 to 23.55	m ³

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Lime stone	—	23.55 to 25.90	m ³
	Marble	—	26.70	m ³
	Pumice	—	7.85 to 11.00	m ³
	Quartz rock	—	25.90	m ³
	Sand stone	—	21.95 to 23.54	m ³
	Slate	—	27.45	m ³
	Soap stone	—	26.45	m ³
51.	Tar, Coal			
	Crude (see IS 212)	—	9.90	m ³
	Naphtha, light	—	9.90	m ³
	Naphtha, heavy	—	9.90	m ³
	Road tar	—	9.90	m ³
	Pitch (see IS 216)	—	9.90	m ³
52.	Thermal Insulation			
	Unbonded glass wool	—	12.75 to 23.55	m ³
	Unbonded rock and slag wool	—	11.30 to 19.60	m ³
	Cellular concrete			
	Grade A	—	Up to 29.40	m ³
	Grade B	—	29.50 to 39.20	m ³
	Grade C	—	39.30 to 49.00	m ³
	Performed calcium silicate Insulation (for temperature up to 650 °C)	—	19.60 to 34.30	m ³
53.	Tiles and Stone Cladding			
	Terra-cotta	—	18.35 to 23.25	m ²
	Terrozzo			
	Paving	10	0.24	m ²
	Cast partitions	40	0.93	m ²
	Tiles			
	Mangalore pattern (see IS 654)	—	0.02 to 0.03	Tile
	Polystyrene wall tiles	99 × 99	0.013	m ²
		148.5 × 148.5	0.013	m ²
	Pressed ceramic tiles, glazed and unglazed (see IS 15622)		22	m ³
	Concrete paving blocks (see IS 15658)		24	m ³
54.	Timber			
	Typical Indian timbers (see IS 399)			
	Aglaia	—	8.34	m ³
	Aini	—	5.83	m ³
	Alder	—	3.63	m ³
	Amari	—	6.13	m ³
	Amla	—	7.85	m ³
	Amra	—	4.41	m ³
	Anjan	—	8.33	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Arjun	—	7.99	m ³
	Ash	—	7.06	m ³
	Axlewood	—	8.82	m ³
	Babul	—	7.70	m ³
	Baen	—	7.70	m ³
	Bahera	—	7.99	m ³
	Bakota	—	4.21	m ³
	Balasu	—	7.55	m ³
	Ballagi	—	11.13	m ³
	Bamboo			
	Banati	—	4.41	m ³
	Benteak	—	6.62	m ³
	Ber	—	6.91	m ³
	Bhendi	—	7.55	m ³
	Bijasal	—	7.85	m ³
	Black chuglam	—	6.13	m ³
	Birch	—	7.85	m ³
	Black locust	—	8.34	m ³
	Blue gum	—	8.34	m ³
	Blue pine	—	5.05	m ³
	Bola	—	6.42	m ³
	Bonsum	—	5.20	m ³
	Bullet wood	—	8.78	m ³
	Casuarina	—	8.34	m ³
	Cettis	—	6.42	m ³
	Champ	—	4.85	m ³
	Chaplash	—	5.05	m ³
	Chatian	—	4.07	m ³
	Chikrassy	—	6.62	m ³
	Chilauni	—	6.42	m ³
	Chilla	—	7.85	m ³
	Chir	—	5.64	m ³
	Chuglam:			
	Black	—	7.85	m ³
	White (silver grey-wood)	—	6.91	m ³
	Cinnamon	—	6.42	m ³
	Cypress	—	5.05	m ³
	Debdaru	—	6.28	m ³
	Deodar	—	5.35	m ³
	Devdam	—	7.06	m ³
	Dhaman:			
	Grewia tiliofolia	—	7.70	m ³
	Grewia vestita	—	7.40	m ³
	Dhup	—	6.42	m ³
	Dilenia	—	6.13	m ³
	Dudhi	—	5.49	m ³
	Ebony	—	8.19	m ³
	Elim	—	5.20	m ³
	Eucalyptus	—	8.33	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Figs	—	4.56	m ³
	Fir	—	4.14	m ³
	Frash	—	6.62	m ³
	Gamari	—	5.05	m ³
	Gardenia	—	7.40	m ³
	Garuga	—	5.98	m ³
	Geon	—	4.07	m ³
	Gluta	—	7.06	m ³
	Gokul	—	4.07	m ³
	Grewia sp.	—	7.55	m ³
	Gurjan	—	7.70	m ³
	Gutel	—	4.41	m ³
	Haldu	—	6.62	m ³
	Hathipaila	—	5.84	m ³
	Hiwar	—	7.70	m ³
	Hollock	—	5.98	m ³
	Hollong	—	7.21	m ³
	Hoom	—	7.21	m ³
	Horse chestnut	—	5.05	m ³
	Imli	—	8.97	m ³
	Indian Chestnut	—	6.28	m ³
	Indian Hemlock	—	3.92	m ³
	Indian Oak	—	8.48	m ³
	Indian Olive	—	10.35	m ³
	Irul	—	8.33	m ³
	Jack	—	5.83	m ³
	Jaman	—	7.70	m ³
	Jarul	—	6.13	m ³
	Jathtkai	—	5.05	m ³
	Jhingan	—	5.63	m ³
	Jutili	—	7.85	m ³
	Kadam	—	4.85	m ³
	Kail	—	5.05	m ³
	Kaim	—	6.42	m ³
	Kambli	—	4.07	m ³
	Kanchan	—	6.62	m ³
	Kanjuh	—	5.84	m ³
	Karada	—	8.34	m ³
	Karal	—	7.99	m ³
	Karani	—	6.28	m ³
	Karar	—	5.34	m ³
	Kardahi	—	9.27	m ³
	Karimgotta	—	3.92	m ³
	Kasi	—	5.83	m ³
	Kasum	—	10.84	m ³
	Kathal	—	5.85	m ³
	Keora	—	6.13	m ³
	Khair	—	9.00	m ³
	Khasipine	—	5.05	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Kindal	—	7.55	m ³
	Kokko	—	6.28	m ³
	Kongoo	—	9.76	m ³
	Kuchla	—	8.63	m ³
	Kumbi	—	7.70	m ³
	Kurchi	—	5.20	m ³
	Kurung	—	9.76	m ³
	Kusum	—	11.28	m ³
	Kuthan	—	4.71	m ³
	Lakooch	—	6.28	m ³
	Lambapatti	—	5.34	m ³
	Lampati	—	5.05	m ³
	Laurel	—	8.33	m ³
	Lendi	—	7.40	m ³
	Machilus:			
	Gamblei	—	5.05	m ³
	Macrantha	—	5.20	m ³
	Maharukh	—	4.07	m ³
	Mahogany	—	6.62	m ³
	Mahua	—	8.97	m ³
	Maina	—	5.64	m ³
	Makai	—	3.14	m ³
	Malabar neem	—	4.41	m ³
	Mango	—	6.77	m ³
	Maniawga	—	7.40	m ³
	Maple	—	5.64	m ³
	Mesua	—	9.76	m ³
	Milla	—	9.12	m ³
	Mokha	—	7.99	m ³
	Mulberry	—	6.62	m ³
	Mullilam	—	7.21	m ³
	Mundani	—	6.77	m ³
	Murtenga	—	7.70	m ³
	Myrabolan	—	9.27	m ³
	Narikel	—	5.49	m ³
	Nedunar	—	5.05	m ³
	Oak	—	8.48	m ³
	Padauk	—	7.06	m ³
	Padri	—	7.06	m ³
	Palang	—	5.98	m ³
	Pali	—	6.28	m ³
	Papita	—	3.28	m ³
	Parrotia	—	8.48	m ³
	Persian lilac	—	5.84	m ³
	Piney	—	6.13	m ³
	Ping	—	8.97	m ³
	Pinus insignis	—	6.13	m ³
	Pipli	—	5.83	m ³
	Pitraj	—	6.77	m ³

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Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
	Poon	—	6.42	m ³
	Poplar	—	4.41	m ³
	Pula	—	3.78	m ³
	Pyinma	—	3.98	m ³
	Rajbrikh	—	8.48	m ³
	Red sanders	—	10.84	m ³
	Rohini	—	11.33	m ³
	Rosewood (black wood)	—	8.19	m ³
	Rudrak	—	4.71	m ³
	Sal	—	8.48	m ³
	Salai	—	5.64	m ³
	Sandal wood	—	8.97	m ³
	Sadan	—	8.34	m ³
	Satin wood	—	9.41	m ³
	Saykaranji	—	7.40	m ³
	Seleng	—	4.85	m ³
	Semul	—	3.78	m ³
	Silver oak	—	6.28	m ³
	Siris	—	3.92	m ³
	Kala-siris	—	7.21	m ³
	Safed- siris	—	6.28	m ³
	Siaso	—	7.70	m ³
	Spruce	—	4.71	m ³
	Suji	—	2.65	m ³
	Sundri	—	9.41	m ³
	Talauma	—	5.64	m ³
	Tanaku	—	2.09	m ³
	Teak	—	6.28	m ³
	Toon	—	5.05	m ³
	Udal	—	2.50	m ³
	Upas	—	3.14	m ³
	Uriam	—	7.40	m ³
	Vakai	—	9.41	m ³
	Vellapine	—	5.83	m ³
	Walnut	—	5.64	m ³
	White bombwe	—	5.98	m ³
	White cedar	—	7.06	m ³
	White chuglam (silver grey-wood)	—	6.91	m ³
	White dhup	—	4.22	m ³
	Yon	—	8.33	m ³
NOTE — The unit of timbers correspond to average unit of typical Indian timbers at 12 percent moisture content.				

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SI No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
55.	Water			
	Fresh	—	10.0	m ³
	Salt	—	10.05	m ³
56.	Wood – Wool Building Slabs	10mm	0.059	m ²
57.	Walling (see IS 6072)			
	Autoclaved reinforced cellular concrete wall slabs			
	Class A	—	8.35 to 9.80	m ²
	Class B	—	7.35 to 8.35	m ²
	Class C	—	6.35 to 7.35	m ²
	Class D	—	5.40 to 6.35	m ²
	Class E	—	4.40 to 5.40	m ²
	Brick masonry (see 38 'Masonry brick' in this Table)			
	Blocks masonry (see 11 'Block' in this Table)			
	Stone masonry (see 37 'Masonry stone' in this Table)			
	Partitions			
	Brick wall	100	1.91	m ²
	Cinder concrete	75	1.13	m ²
	Galvanized on sheet	—	0.15	m ²
	Hollow glass block (bricks)	100	0.88	m ²
	Hollow blocks per 200 mm of thickness			
	Ballast or stone concrete	200	0.201	m ²
	Clay	200	0.201	m ²
	Clinker concrete	200	0.220	m ²
	Coke breeze concrete	200	9.176	m ²
	Diatomaceous earth	200	0.093	m ²
	Gypsum	200	0.137	m ²
	Pumice concrete	200	0.177	m ²
	Slag concrete, air-cooled	200	0.196	m ²
	Slag concrete foamed	200	0.186	m ²
	Lath and plaster	—	0.392	m ²
	Solid blocks per 20 mm of thickness			
	Ballast or stone	20	0.451	m ²
	Clinker concrete	20	0.300	m ²
	Coke breeze concrete	20	0.221	m ²
	Pumice concrete	20	0.221	m ²
	Slag concrete, foamed	20	0.250	m ²
	Terrazzo cast partitions	40	0.932	m ²
	Timber studding plastered	—	9.981	m ²
	NOTE — For unit weight of fixtures and fittings required to buildings including builder's hardware, reference may be made to appropriate Indian standards.			

Table 2 Unit Weights of Stored and Miscellaneous Materials

(Clauses [4.1](#) and [4.2](#))

Sl No.	Materials	Weight KN/m ³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
1.	Agriculture and Food Products		
	Butter	8.45	–
	Coffee in bags	5.50	–
	Drinks in bottle, in boxes	7.35	–
	Eggs, packed	2.95	–
	Fats, oil	5.80	–
	Fish meal	4.90	45
	Flour in sacks up to 1 m height	2.20 to 5.90	–
	Forage (bales)	1.25	–
	Fruits	3.45	–
	Grains:		
	Barley	6.75	27
	Corn, shelled	7.55	27
	Flax seed	7.35	30
	Oats	5.30	30
	Rice	6.55	33
	Soyabeans	7.35	30
	Wheat	8.15	28
	Wheat flour	6.85	30
	Grain sheaves up to 4 m stack height	0.98	30
	Grain sheaves over 4 m stack height	1.45	30
	Grass and clover	3.45	–
	Hay:		
	Compressed	1.65	–
	Loose up to about 3 m stack height	0.69	–
	Honey	14.10	–
	Hops:		
	In stacks	1.65	–
	In cylinder hop bins	4.60	–
	Sewn up or compressed in cylindrical shape in hop cloth	2.85	–
	Malt:		
	Crushed	3.90	20
	Germinated	1.85	–
	Meat and meat products	7.05	–
	Milk	10.05	–
	Molasses	4.40	–
	Onion in bags	5.40	0

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Sl No.	Materials	Weight KN/m ³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Oil cakes, crushed	5.80	0
	Potatoes	7.05	30
	Preserves (tins in cases)	4.90 to 7.85	—
	Bags	7.05	—
	Bulk	9.40	—
	Seeds:		
	Heaps	4.90 to 7.85	25
	Sacks	3.90 to 6.85	—
	Straw and chaff:		
	Loose up to about 3 m stack height	0.45	—
	Compressed	1.65	—
	Sugar:		
	Crystal	7.35	30
	Cube sugar in boxes	7.85	—
	Sugar beet, pressed out	7.85	—
	Tobacco bundles	3.45	—
	Vinegar	10.40	—
	2. Chemicals and Allied Materials		
	Acid, hydrochloric	11.75	—
	Acid, nitric 91 percent	14.80	—
	Acid, sulphuric 87 percent	17.55	—
	Alcohol	7.65	—
	Alum, pearl in barrel	5.20	—
	Ammonia, liquid	8.85	—
	Ammonium, chloride, crystalline	8.15	30 to 40
	Ammonium nitrate	7.05 to 9.80	25
	Ammonium sulphate	7.05 to 9.00	32 to 45
	Beeswax	9.40	—
	Benzole	8.90	—
	Benzene hexachloride	8.75	—
	Bicarbonate of soda	6.40	—
	Bone	18.65	—
	Borax	17.15	—
	Calcite	26.50	—
	Camphor	9.70	—
	Carbon disulphide	12.75	—
	Casein	13.25	—
	Caustic soda	13.85	—
	Creosole	10.50	—
	Dicalcium phosphate	6.65	—

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Sl No.	Materials	Weight KN/m³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Disodium phosphate	3.90 to 4.80	30 to 45
	Iodine	48.55	–
	Oils in bottle or barrels	5.70 to 8.90	–
	Oil, linseed:		
	In barrels	5.70	–
	In drums	7.05	–
	Oil, turpentine	8.50	–
	Paints	9.40	–
	Paraffin wax	7.85 to 9.40	–
	Petroleum	9.90	–
	Phosphorus	17.85	–
	Plastics and polymers:		
	Cellulose acetate	12.25 to 13.35	–
	Cellulose nitrate	13.25 to 15.70	–
	Methyl methacrylate (IS 14753)	11.57 to 11.77	–
	Phenol formaldehyde	12.55	–
	Expanded Polystyrene (IS 4671)	0.15 to 0.35	–
	Extruded polystyrene	0.21	
	Polyvinyl chloride (Perspex)	11.75 to 13.25	–
	Resin bonded sheet	12.85 to 13.55	–
	Urea formaldehyde	13.25 to 13.55	–
	Extruded polystyrene	0.21	
	Polyurethane	0.59 to 0.66	
	Polyethylene (IS 2508)	9.81 to 19.6	
	Polyvinyl Butaryl	11	
	Polycarbonate (IS 14443)	12	
	Ethylene tetrafluoroethylene	17	
	Polytetrafluoroethylene [IS 14635 (part 2)]	22	
	Potash	14.40	–
	Potassium	8.65	–
	Potassium nitrate	9.90	–
	Red lead, dry	20.70	–
	Red lead, paste	87.30	–
	Rosin in barrels	6.75	–
	Rubber:		
	Raw	8.90 to 9.40	–
	Vulcanized	8.90 to 9.10	–
	Saltpetre	9.91	–
	Sodium silicate in barrels	8.35	–
	Sulphur	20.10	–

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Sl No.	Materials	Weight KN/m ³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Talc	27.45	–
	Varnishes	9.40	–
	Vitriol, blue, in barrels	7.05	–
3.	Fuels		
	Brown coal	6.83	–
	Brown coal briquettes heaped	7.85	35
	Brown coal briquettes stacked	12.75	–
	Charcoal	2.95	–
	Coal:		
	Untreated, mine-moist	9.80	35
	In washeries	11.75	0
	Dust	6.85	25
	All other sorts	8.35	35
	Coke:		
	Furnace or gas	4.90	35
	Brown coal, low-temperature	9.80	35
	Hard, raw coal	8.35	35
	Hard, raw coal, mine-damp	9.80	35
	Diesel oil	9.40	0
	Firewood, chopped	3.90	45
	Petrol	6.75	0
	Wood in chips	1.95	45
	Wood shaving, loose	1.45	35
	Wood shaving, shaken down	2.45	35
4.	Manures		
	Animal manures:		
	Loosely heaped	11.75	45
	Stacked dung up to 25 m stack height	17.65	45
	Artificial manures	11.75	24 to 30
5.	Metals and Alloys		
	Aluminium	25.30 to 26.60	
	Cast	25.30 to 26.60	–
	Wrought	25.90 to 27.45	–
	Sheet per mm of thickness per m ²	0.028	–
	Antimony, pure:		
	Amorphous	60.90	–
	Solid	65.70	–
	Bismuth:		
	Liquid	98.07	–
	Solid	95.02 to 97.9	–

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Sl No.	Materials	Weight KN/m³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Cadmium:		
	Cast	83.75 to 84.05	—
	Wrought	85.03	—
	Calcium	15.60	—
	Chromium	63.95 to 66.00	—
	Cobalt:		
	Cast	83.25 to 85.10	—
	Wrought	88.45	—
	Copper:		
	Cast	86.20 to 87.65	—
	Wrought	86.70 to 87.65	—
	Sheet per mm of thickness	0.09	—
	Gold:		
	Cast	188.75 to 189.55	—
	Wrought	189.55	—
	Iron:		
	Pig	70.60	—
	Grey, cast	68.95 to 69.90	—
	White, cast	74.35 to 75.70	—
	Wrought	75.50	—
	Lead:		
	Cast	111.20	—
	Liquid	105.00	—
	Wrought	111.40	—
	Sheet per mm of thickness	0.11	—
	Magnesium	16.45 to 17.15	—
	Manganese	72.55	—
	Mercury	133.35	—
	Nickel	81.20 to 87.20	—
	Platinum	210.25	—
	Silver:		
	Cast	102.0 to 102.85	—
	Liquid	93.15	—
	Wrought	103.35 to 103.55	—
	Sodium:		
	Liquid	9.10	—
	Solid	9.30	—
	Tungsten	188.30	—
	Uranium	180.45	—
	Zinc		

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Sl No.	Materials	Weight KN/m ³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Cast	68.95 to 70.20	—
	Wrought	70.50	—
	Sheet per mm of thickness	0.07	—
	Alloys:		
	Aluminium and copper		
	Aluminium 10 percent, copper 90 percent	75.40	—
	Aluminium 5 percent, copper 95 percent	82.00	—
	Aluminium 3 percent, copper 97 percent	85.10	—
	Aluminium 91 percent zinc 9 percent	27.45	—
	Babbitt metal (tin 90 percent, lead 5 percent, copper 5 percent)	71.70	—
	Wood's metal (bismuth 50 percent, lead 25 percent, cadmium 12.5 percent, tin 12.5 percent)	95.00	—
	Brasses:		
	Muntz metal (copper 60 percent, zinc 40 percent)	80.60	—
	Red (copper 90 percent, zinc 10 percent)	84.25	—
	White (copper 50 percent, zinc 50 percent)	80.30	—
	Yellow (copper 70 percent, zinc 30 percent)		
	Cast	82.75	—
	Drawn	85.10	—
	Rolled	83.85	—
	Bronzes:		
	Bell metal (copper 80 percent, tin 20 percent)	85.60	—
	Gun metal (copper 90 percent, tin 10 percent)	86.10	—
	Cadmium and tin	75.40	—
	German silver:		
	Copper 52 percent, zinc 26 percent, nickel 22 percent	82.75	—
	Copper 59 percent, zinc 30 percent, nickel 11 percent	81.70	—
	Copper 63 percent, zinc 30 percent, nickel 7 percent	81.40	—
	Gold and Copper:		
	Gold 98 percent, copper 2 percent	184.75	—
	Gold 90 percent, copper 10 percent	168.20	—
	Lead and Tin:		
	Lead 87.5 percent, tin 12.5 percent	103.85	—

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Sl No.	Materials	Weight KN/m ³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Lead 30.5 percent, tin 69.5 percent	81.10	–
	Monel metal cast (nickel 70 percent, copper 30 percent)	87.00	–
	Steel:		
	Cast	77.00	–
	Wrought mild	76.30	–
	Black plate per mm of thickness	0.08	–
	Steel sections (<i>see</i> 46 ‘Steel sections’ in this Table)		
6.	Miscellaneous Materials		
	Aggregate, coarse	10.80 to 15.70	30
	Ashes, coal, dry, 12 mm and under	5.50 to 6.30	40
	Ashes, coal, dry, 75 mm and under	5.50 to 6.30	38
	Ashes, coal, wet 12 mm and under	7.05 to 7.85	52
	Ashes, coal, wet 75 mm and under	7.05 to 7.85	50
	Asphalt, crushed 12 mm and under	7.05	30 to 45
	Ammonium nitrate, polls	3.55 to 8.35	27
	Bone	18.65	–
	Books and files, stacked	8.35	–
	Calcium ammonium nitrate	9.80	28
	Copper sulphate, ground	11.75	30
	Chalk	21.95	–
	Chinaware, earthenware, stacked (including cavities)	10.80	–
	Clinker, furnace, clean	7.85	30
	Diammonium phosphate	7.85 to 8.50	29
	Double salt (ammonium sulphate nitrate)	7.05 to 9.30	34
	Filling cabinet and cupboards with contents in records offices, libraries, archives	5.90	–
	Flue dust, boiler house, dry	5.50 to 7.05	30
	Fly ash, pulverised	5.50 to 7.05	–
	Glass, solid	23.50 to 26.70	–
	Wool	0.16 to 1.18	–
	In sheets	25.50	–
	Glue	12.55	–
	Gypsum, calcined, 12mm and under	8.60 to 9.40	40
	Gypsum, calcined, powdered	9.40 to 12.55	45
	Gypsum, raw, 25 mm and under	14.10 to 15.70	30 to 45
	Hides:		
	Dry	8.65	–
	Salted } Only green	8.90	–

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Sl No.	Materials	Weight KN/m³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Ice		
	Leather put in rows	7.85	–
	Lime, ground, 3 mm and under	9.40	> 45
	Lime, hydrated, 3 mm and under	6.30	30 to 45
	Lime, hydrated, pulverized	5.00 to 6.30	30 to 45
	Lime pebble	8.25 to 8.75	
	Limestone, agricultural, 3 mm and under	10.60	
	Limestone, crushed	13.30 to 14.10	30 to 45
	Limestone dust	8.65 to 14.90	30 to 45
	Magnesite, caustic, in powder form	7.85	–
	Magnesite sinter and Magnesite granular	19.60	–
	Phosphate, rock, pulverized	9.40	40 to 52
	Phosphate rock	11.75 to 13.35	30 to 45
	Phosphate sand	14.10 to 15.70	30 to 45
	Potassium carbonate	7.95	30 to 45
	Potassium chloride, pellets	18.85 to 20.40	30 to 45
	Potassium nitrate	4.85	> 30
	Potassium sulphate	6.55 to 7.45	45
	Pyrites, pellets	18.85 to 20.40	30 to 45
	Pumice	5.80 to 9.90	–
	Rubbish:		
	Building	13.80	–
	General	6.30	–
	Salt, common, dry, coarse	6.30 to 10.00	30 to 45
	Salt, common, dry, fine	11.00 to 12.55	30 to 45
	Salt cake, dry, coarse	13.35	30
	Salt cake, dry, pulverized	11.20 to 13.35	35
	Sand, bank, damp	17.25 to 20.40	45
	Sand, bank, dry	14.10 to 17.25	30
	Sand, silica, dry	14.10 to 15.70	30 to 45
	Saw dust, loose	1.57	30
	Silica gel	4.40	30 to 45
	Soda ash, heavy	8.67 to 10.20	35
	Soda ash, light	4.70 to 6.00	37
	Sodium nitrate, granular	11.00 to 12.55	24
	Sulphur, crushed, 12 mm and under	7.85 to 8.25	35 to 45
	Sulphur, 76 mm and under	8.65 to 13.35	32
	Sulphur, powdered	7.85 to 9.40	30 to 45
	Single superphosphate (S.S.P), granulated	7.65 to 8.25	37
	Slag, furnace, crushed	14.90	35

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Sl No.	Materials	Weight KN/m³	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Steel goods:		
	Cylinders, usually stored for Carbonic acid, etc	13.80	
	Sheets, railway rails, etc, usually stored	44.00	
	Trisodium phosphate	9.40	30 to 45
	Triple superphosphate	7.85 to 8.65	30 to 45
	Turf	2.85 to 5.70	–
	Urea, prills	6.40	23 to 26
7.	Ores		
	Antimony	29.80	–
	Ferrous sulphide	26.50	–
	Ferrous sulphide ore waste after roasting	13.85	–
	Iron ore, compact storing	29.80	–
	Magnesium ore	19.60	–
8.	Textiles, Paper and Allied Materials		
	Cellulose in bundles	7.35	–
	Cotton, compressed	12.75	–
	Flax, piles and compressed in bales	2.95	–
	Furs	8.80	–
	Jute in bundles	6.85	–
	Paper:		
	In bundles and rolls	6.85	–
	Newspaper in bundles	3.90	–
	Put in rows	10.80	–
	Thread in bundles	4.90	–
	Wood, compressed	12.75	–

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ANNEX A

(Clause 2)

LIST OF REFERRED STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
IS 212 : 2021	Crude coal tar for general use (<i>third revision</i>)	IS 654 : 2025	Clay roofing tiles mangalore Pattern — Specification (<i>fifth revision</i>) under revision
IS 216 : 2006	Coal tar pitch — Specification (<i>second revision</i>)	IS 808 : 2021	Hot rolled steel beam, column, channel and angle sections — Dimensions and properties (<i>fourth revision</i>)
IS 269 : 2015	Ordinary Portland Cement — Specification (<i>sixth revision</i>)	IS 809 : 1992	Rubber flooring materials for general purposes — Specification (<i>first revision</i>)
IS 277 : 2018	Galvanized steel strips and sheets (Plain and Corrugated) — Specification (<i>seventh revision</i>)	IS 811 : 1987	Specification for cold formed light gauge structural steel sections (<i>second revision</i>)
IS 303 : 2024	Plywood for General Purposes — Specification (<i>fourth revision</i>)	IS 1077 : 2025	Common burnt clay building bricks — Specification (<i>sixth revision</i>)
IS 383 : 2025	Coarse and fine aggregate for concrete — Specification (<i>fourth revision</i>) under revision	IS 1173 : 1978	Specification for hot rolled and slit steel tee bars (<i>second revision</i>)
IS 399 : 1963	Classification of commercial timbers and their zonal distribution (<i>revised</i>)	IS 1322 : 1993	Bitumen Felts for Water Proofing and Damp-proofing — Specification (<i>fourth revision</i>)
IS 404 (Part1) : 1993	Lead pipes — Specification: Part 1 For other than chemical purposes (<i>third revision</i>)	IS 1343 : 2012	Prestressed concrete — Code of Practice (<i>second revision</i>)
IS 412 : 1975	Specification for expanded metal steel sheets for general purposes (<i>second revision</i>)	IS 1478 : 2023	Clay flooring tiles specification (<i>third revision</i>)
IS 457 : 1957	Code of practice for general construction of plain and reinforced concrete for dams and other massive structures	IS 1536 : 2023	Centrifugally cast (Spun) iron pressure pipes for water, gas and sewage — Specification (<i>fifth revision</i>)
IS 458 : 2021	Precast concrete pipes (with and without reinforcement) — Specification (<i>fifth revision</i>)	IS 1537 : 1976	Specification for vertically cast iron pressure pipes for water, gas and sewage (<i>first revision</i>)
IS 459 : 1992	Corrugated and semi- corrugated asbestos cement sheets — Specification (<i>third revision</i>)	IS 1592 : 2003	Asbestos cement pressure pipes and joints — Specification (<i>fourth revision</i>)
IS 651 : 2007	Glazed stoneware pipes and fittings — Specification (<i>sixth revision</i>)	IS 1626	Asbestos cement building pipes and pipe fittings, gutters and gutter fittings and roofing fittings — Specification:
IS 653 : 1992	Linoleum sheets and tiles — Specification (<i>third revision</i>)		

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<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
(Part 1) : 1994	Pipe and pipe fittings (<i>second revision</i>)	(Part 2) : 2023	Part 2 Cold Formed Sheet Pile (<i>second revision</i>)
(Part 2) : 1994	Gutter and gutter fittings (<i>second revision</i>)	IS 2394 : 1984	Code of Practice For Application of Lime Plaster Finish (<i>first revision</i>)
IS 1658 : 2006	Fibre hardboards — Specification (<i>third revision</i>)	IS 2508 : 2024	Polyethylene films and sheets — Specification (<i>fourth revision</i>)
IS 1725 : 2023	Stabilized soil blocks used in general building construction — Specification (<i>third revision</i>)	IS 2526 : 1963	Code of practice for acoustical design of auditoriums and conference halls
IS 1726 : 1991	Cast iron manhole covers and frames — Specification (<i>third revision</i>)	IS 2541 : 1991	Preparation and use of lime concrete — Code of practice (<i>second revision</i>)
IS 1729 : 2023	Sand cast iron spigot and socket pipes, fittings and accessories — Specification (<i>third revision</i>)	IS 2547	Specification for gypsum building plaster
IS 2095	Gypsum plaster boards — Specification:	(Part 1) : 1976	Excluding premixed lightweight plasters (<i>first revision</i>)
(Part 1) : 2011	Plain gypsum plaster boards (<i>fourth revision</i>)	(Part 2) : 1976	Premixed lightweight plasters (<i>first revision</i>)
(Part 2) : 2022	Coated laminated gypsum plaster boards (<i>third revision</i>)	IS 2553 (Part 1) : 2018	Safety glass — Specification: Part 1 architectural, building and general uses (<i>fourth revision</i>)
(Part 3) : 2022	Reinforced gypsum plaster boards and ceiling tiles (<i>fourth revision</i>)	IS 2835 : 1987	Specification for flat transparent sheet glass (<i>third revision</i>)
IS 2180 : 2025	Specification for heavy duty burnt clay building bricks (<i>fourth revision</i>) under revision	IS 3087 : 2005	Particle boards of wood and other lignocellulosic materials (medium density) for general purposes — Specification (<i>second revision</i>)
IS 2185	Concrete masonry units — Specification	IS 3115 : 1992	Lime based blocks — Specification (<i>second revision</i>)
(Part 1) : 2005	Part 1 hollow and solid concrete blocks (<i>third revision</i>)	IS 3129 : 1985	Specification for low density particle boards
(Part 3) : 1984	Part 3 Autoclaved Cellular Aerated Concrete Blocks	IS 3348 : 1965	Specification for fibre insulation boards
IS 2222 : 2025	Specification for burnt clay perforated building bricks (<i>fourth revision</i>) under revision	IS 3478 : 1966	Specification for high density wood particle boards
IS IS 2314	Steel Sheet Piling Section — Specification	IS 4139: 1989	Calcium Silicate Bricks — Specification (<i>second revision</i>)
(Part 1) : 2023	Part 1 Hot Rolled Sheet Pile (<i>second revision</i>)		

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<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
IS 4253	Cork composition sheets — Specification		lignocellulosic material — Specification (<i>first revision</i>)
(Part 1) : 2008	Plain cork sheets (<i>second revision</i>)	IS 12894 : 2002	Pulverized fuel ash — Lime bricks — Specification (<i>first revision</i>)
(Part 2) : 2008	Cork and rubber (<i>second revision</i>)	IS 14443 : 1997	Polycarbonate sheets — Specification
IS 4671 : 2018	Expanded polystyrene for thermal insulation purposes (<i>second revision</i>)	IS 14587 : 2023	Prelaminated medium density fibre board — Specification
IS 486 0 : 1968	Specification for acid — Resistant bricks	IS 14635	Fluoropolymer dispersions and moulding and extrusion materials
IS 4923 : 2017	Hollow steel sections for structural use — Specification (<i>third revision</i>)	(Part 1) : 2020	Designation system (<i>first revision</i>)
IS 6072 : 2023	Autoclaved reinforced cellular concrete wall slabs — Specifications (<i>first revision</i>)	(Part 2) : 2020	Preparation of test specimens and determination of properties (<i>first revision</i>)
IS 6598 : 2018	Cellular concrete for thermal insulation - Specification (<i>first revision</i>)	IS 14753 : 1999	Polymethyl methacrylate (pmma) (acrylic) sheets
IS 8041 : 1990	Rapid hardening portland cement — Specification (<i>second revision</i>)	IS 14862 : 2000	Fibre cement flat sheets — specification
IS 9012 : 1978	Recommended practice for shotcreting	IS 15622 : 2017	Pressed ceramic tiles — Specification (<i>first revision</i>)
IS 12679 : 2023	By-product gypsum for construction — Specification (<i>second revision</i>)	IS 15658 : 2001	Concrete paving blocks — Specification (<i>first revision</i>)
IS 12823 : 2015	Prelaminated particle boards from wood and other	IS 16720 : 2018	Pulverized fuel ash-cement bricks — Specification
		IS 17452 : 2020	Use of alkali activated concrete for precast products — Guidelines

ANNEX B

(Foreword)

COMMITTEE COMPOSITION

Structural Safety Sectional Committee, CED 37

<i>Organization</i>	<i>Representative(s)</i>
Indian Institute of Technology Bombay, Mumbai	DR RAVI SINHA (Chairperson)
Atomic Energy Regulatory Board, Mumbai	DR A. D. ROSHAN
Bharat Heavy Electrical Limited, New Delhi	SHRI FELIX ANTONY SHRI MAHENDRA KUMAR NAHAK (<i>Alternate</i>)
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Indian Institute of Technology, Indore	DR LALIT BORANA
MECON Limited, Ranchi	SHRI R. S. RAMAN SHRI ARIJIT NATH (<i>Alternate</i>)

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National Council for Cement and Building Materials, Faridabad	SHRI P. N. OJHA SHRI BRIJESH SINGH (<i>Alternate</i>)
National Institute of Technology, Rourkela	PROF PRADIP SARKAR
RWDI Consulting Engineers (India) Private Limited, Trivandrum	DR K. SURESH KUMAR SHRI RAHUL P. S. (<i>Alternate</i>)
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IS 875 (Part 1) : 2026

Panel for Revision of IS 875 (Part 1)

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CSIR – Central Building Research Institute, Roorkee	SHRI SIDDHARTH BEHERA
CSIR – Structural Engineering Research Centre, Chennai	DR G. V. RAMA RAO
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also been considered in this revision. Provisions relating to dead load for gardening and landscaping have also been introduced.

This standard contributes to the Sustainable Development Goal 9 ‘Build resilient infrastructure, promote sustainable industrialization and foster innovation’.

The composition of the Committee responsible for the revision of this standard is given in [Annex B](#).

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be periodically removed to create more space for the future falling blocks.

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Amendments Issued Since Publication

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